4 Day Work Week Statistical Summary

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OSY/315: Statistical Reasoning in Psychology

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06/17/2024

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The problem we are addressing is whether implementing a four-day workweek in a large tech company will increase employee motivation and subsequently improve performance and productivity. The hypothesis will be tested in two departments before considering broader organizational changes. To evaluate this, the Devine Company employee satisfaction surveys will be leveraged to measure changes in motivation and performance.

The idea of a four-day workweek has gathered significant attention in recent years, driven by concerns about employee productivity and engagement. Like our scenario, the founder and director of a medium-sized company found an article in the article in the Economist that reported office employees are only productive for 1.5–2 hours during an 8-hour workday. This observation led to an 8-week trial to reduce work hours from 4 days (40 hours) to a four-day (32 hours) work week. This trial had been a "resounding success," and a four-day work week was implemented following the 8-week trial (Delaney & Casey, 2021).

Research has shown that reduced work hours can lead to several benefits, including increased productivity, improved work-life balance, and heightened employee engagement. For instance, a study by Kelliher and Anderson (2010) demonstrated that flexible working arrangements, including reduced work hours, can lead to improved employee motivation and performance.

The research findings provide valuable insights into how reduced work hours can impact productivity and employee engagement. The primary motivation for reducing work hours in the New Zealand study was to enhance productivity and engagement. Employees were encouraged to focus on their productivity rather than hours worked, aligning personal desires with productivity and efficiency. This approach fostered a sense of empowerment and engagement

among employees, as they felt valued and appreciated by the company. Additionally, the study highlighted the importance of employee involvement in designing and measuring work processes. Teams were responsible for defining their work schedules, productivity measures, and support requirements, which stimulated engagement and fostered a collaborative work environment. Employees reported positive experiences during the trial, noticed increased involvement in personal and family activities, improved mental well-being, and enhanced motivation.

The study also revealed potential challenges, such as increased work intensity and stress levels. Some employees experienced heightened pressure to complete tasks within a shorter timeframe, leading to concerns about work quality and personal well-being. These findings suggest that while a four-day workweek can offer significant benefits and drive motivation to complete tasks, it may have an adverse effect on employee well-being and quality of work.

Research Hypothesis

Based on the background research and findings, the following research hypothesis can be drawn: **Implementing a four-day workweek in two departments of the company will lead to increased employee satisfaction.**

- **Null Hypothesis (H0):** There is no difference in employee motivation between the 5-day workweek (Group 1) and the 4-day workweek (Group 2)
- **Research Hypothesis (H1):** There is a significant different in employee motivation between the 5 day workweek (Group 1) and the 4 day workweek (Group 1).

Since the hypothesis does not specify a direction, a two-tailed test is appropriate. A two-tailed test is used to determine if there is a significant difference in either direction. In this

scenario, the research hypothesis states that there is a significant difference in employee performance between the 5-day workweek and the 4-day workweek. The research hypothesis is non-directional; it predicts a difference in either direction without specifying which group will have higher or lower performance. Using a two-tailed test allows us to detect differences in both directions.

Sampling Method & Limitations

We will use random sampling to select our sample groups. This method ensures that each employee has an equal chance of being chosen for either the 4-day workweek group or the 5-day workweek group. Random sampling is ideal for this experiment because it minimizes bias and ensures the groups are representative of the entire population of employees. The sample size for each group is 30. Group 1 will consist of the 5-day workweek employees. Group 2 will include the 4-day workweek employees. While this sample size is statistically significant, it is relatively small and may not perfectly represent the full employee population.

There are a few key limitations to consider with this sampling method. For instance, a sample size of 30 might not capture the diversity and variability present in the entire employee population. This could affect the generalizability of the findings. Another limitation could be that even if the new work schedule is implemented consistently, the way it is perceived and adopted by different employees may vary, introducing potential biases. Another consideration is that the sample might include employees who are already highly motivated or those who are entirely disengaged. These extremes can skew the results, as highly motivated employees might perform well regardless of the schedule, while disengaged employees might not respond positively to any change.

Including descriptive data such as gender, age, job role, and work schedule will be essential for understanding any nuance impacting the 4-day workweek on different employee groups. This will ensure a comprehensive analysis. For example, gender roles and responsibilities, such as working mothers balancing full-time employment and childcare, can significantly affect how different employees perceive and benefit from a 4-day workweek. Another example could be that younger employees might adapt differently to a 4-day workweek compared to older employees. Collecting descriptive data will provide additional layers of insight for potential obscure variables.

Results

To compare the motivation levels of employees working a four-day and five-day workweek, we used a statistical test called a two-sample t-test. This test helps determine if there's a significant difference between two independent groups, in this case, employees on different work schedules. We set a significance level of 0.005, which means we consider results with a p-value less than 0.05 as indicating a statistically significant difference. The research hypothesis suggested that there would be a significant difference in employee motivation between the two groups, without specifying the direction of this difference. Therefore, we conducted a two-tailed test.

After analyzing the data, we found a statistically significant difference in employee motivation between the two workweek schedules. Employees on a four-day workweek reported significantly higher motivation levels (mean of 7.33) compared to those on a five-day workweek (mean of 5.61). The variance in motivation scores was also lower among employees during the four-day workweek.

The two-sample t-test yielded a p-value of 0.000579, which is much smaller than our significance level of 0.05. This indicates strong evidence that the difference in motivation levels between the two groups is unlikely due to random chance. Therefore, we reject the null hypothesis and conclude that there is indeed a statistically significant difference in employee motivation between the four-day and five-day workweeks.

Summary Table for Reference:

	Group 1 (5-day)	**Group 2 (4-day)**
Mean	5.61	7.33
Variance	4.79	1.85
Observations	30	30
T Stat	-3.645	
P-value	0.000573	
Alpha Level	.05	

Based on these findings, it is recommended to implement a four-day workweek across the organization on a trial basis to assess its effectiveness in enhancing employee motivation and satisfaction. This change has the potential to not only boost motivation but also improve overall employee satisfaction and potentially increase productivity.

It’s recommended to monitor this trial phase closely over a period of six months. If the pilot proves successful in maintaining or enhancing motivation levels without compromising productivity or work quality, full implementation of the four-day workweek should be considered. To optimize the effectiveness of the trial period, employees should have the option to voluntarily opt into the four-day workweek. As we’ve learned from other research, giving employees autonomy in choosing their work schedule contributes significantly to the success of such initiatives.

Future research should focus on investigating the long-term effects of a four-day workweek on employee performance and well-being. Additionally, it would be valuable to study how different demographic groups within the organization adapt to the new schedule. Monitoring the impact on workload intensity and stress levels is also essential to ensure that the transition to a four-day workweek does not inadvertently lead to negative effects on employee well-being.

The findings of this study strongly support the implementation of a four-day workweek as a strategy to enhance employee motivation. By carefully monitoring and addressing potential challenges, organizations can successfully transition to this new work schedule, potentially yielding significant benefits for both employees and the organization as a whole.

References

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